Linear Transformations HW 2

1) $f(x) = 2x+4$, $g(x) = x^2 - 3x$, $h(x) = \dfrac{\sqrt{x}}{x-2}$

a) $f[g(x)] = 2(x^2-3x)+4$
$= \boxed{2x^2-6x+4}$

$g[f(x)] = (2x+4)^2 - 3(2x+4)$
$= 4x^2+8x+8x+16-6x-12$
$= \boxed{4x^2+10x+4}$

b) $f(g(2))$
$g(2) = (2)^2-3(2) = 4-6 = -2$
$f(g(2)) = f(-2) = 2(-2)+4 = \boxed{0}$

$g(f(2))$
$f(2) = 2(2)+4 = 8$
$g(f(2)) = g(8) = (8)^2 - 3(8) = \boxed{40}$

c) $h[f(x)] = \dfrac{\sqrt{2x+4}}{2x+4-2} = \boxed{\dfrac{\sqrt{2x+4}}{2x+2}}$

d) $h(f(0))$
$f(0) = 2(0)+4 = 4$
$h(4) = \dfrac{\sqrt{4}}{4-2} = \dfrac{2}{2} = \boxed{1}$

$h(f(-1))$
$f(-1) = 2(-1)+4 = 2$
$h(2) = \dfrac{\sqrt{2}}{2-2} = \dfrac{\sqrt{2}}{0} = \boxed{\text{undefined}}$

e) $f(g(f(x)))$
$g(f(x)) = 4x^2+10x+4$
$f(g(f(x))) = 2(4x^2+10x+4)+4$
$= 8x^2+20x+8+4$
$= \boxed{8x^2+20x+12}$

f) Answers vary but should probably point out that order matters.

2) a) $\vec{y} = G\vec{x} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & -2 \\ 5 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2+8-1 \\ 0+6+2 \\ 5+0-1 \end{bmatrix} = \boxed{\begin{bmatrix} 9 \\ 8 \\ 4 \end{bmatrix}}$

$F\vec{y} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 0 \\ 0 & 3 & 4 \end{bmatrix}\begin{bmatrix} 9 \\ 8 \\ 4 \end{bmatrix} = \begin{bmatrix} 9+0+8 \\ 18-8+0 \\ 0+24+16 \end{bmatrix} = \boxed{\begin{bmatrix} 17 \\ 10 \\ 40 \end{bmatrix}}$